

The Application Layer

Unit 2-Chapter 2

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The Domain Name System

The DNS resolves high-level human readable names for computers to low-level IP addresses

- DNS name space »
- Domain Resource records »
- Name servers »

Why do we need DNS?

- Too difficult to remember IP addresses and if a server is changed everyone would need to be notified of the new IP address
- Originally one file with names and IP addresses – became too large
- Host name conflicts began to occur
- DNS was developed to solve these problems
- DNS is a hierarchical domain based naming scheme and a database system for implementing it.
- It is used primarily for mapping host names to IP addresses

How DNS works

To map a name to an IP address:

- Application program calls a library procedure called a **resolver**, for example *gethostname*, passing it the name as a parameter.
- The resolver sends a query with the name to a local DNS server, which looks up the name and returns the IP address.
- The query and response are sent as UDP packets.
- Once it has the IP address the host can now establish a TCP connection or send UDP packets.

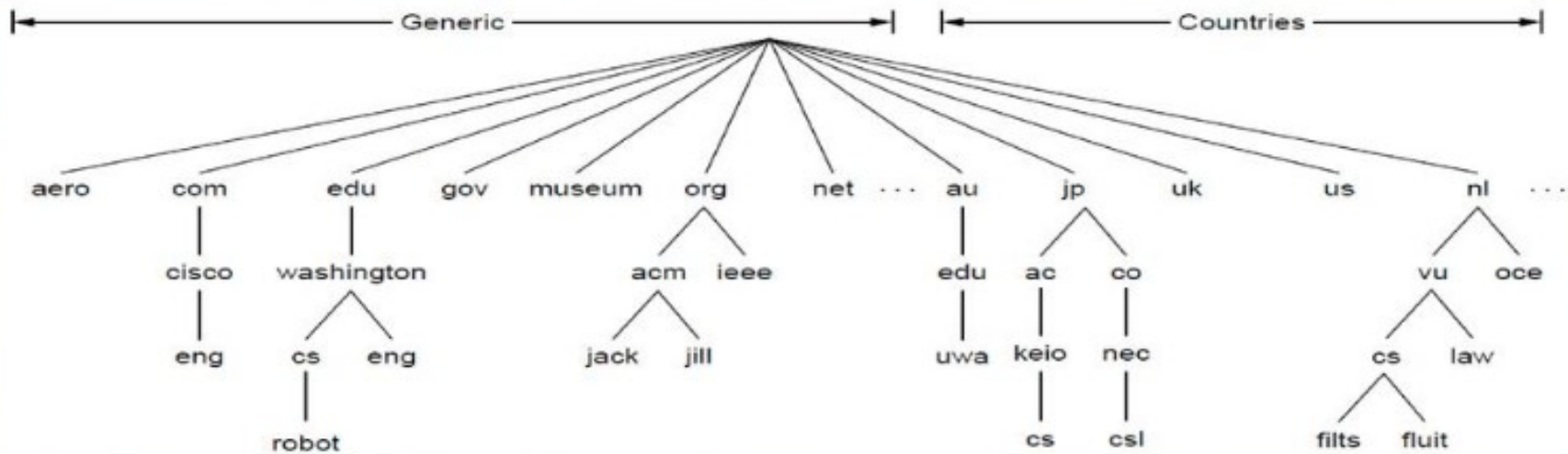
DNS Name Space

- ICANN – (Internet Corporation for Assigned Names and Numbers)
 - Established in 1998 to manage naming hierarchy
 - Internet is divided into 250 top level domains, each partitioned into subdomains
 - Top level domains are Generic (originals from 1980's as well as newer additions) or Countries (one entry per country)
 - Original 1985 domains were com, edu, gov, int, mil net, org
 - Top level domains are run by registrars appointed by ICANN
 - New domains have been added, but often with disagreements (xxx, pro, etc.) as to their use.
 - Domain names are NOT case sensitive
 - Registering a domain with the intent to sell it is called **cybersquatting**

The DNS Name Space

DNS namespace is hierarchical from the root down

- Different parts delegated to different organizations



The computer *robot.cs.washington.edu*

Domain Resource Records

- Domains have a set of resource records associates with them
- A resource record is a five tuple:
domain_name time_to_live class type value
- TTL – large number indicates that this is a stable record
- Class – for internet information – always IN
- Type – what kind of record (see next slide)
 - Most important type is the Address type
- Value – a number, domain name or string

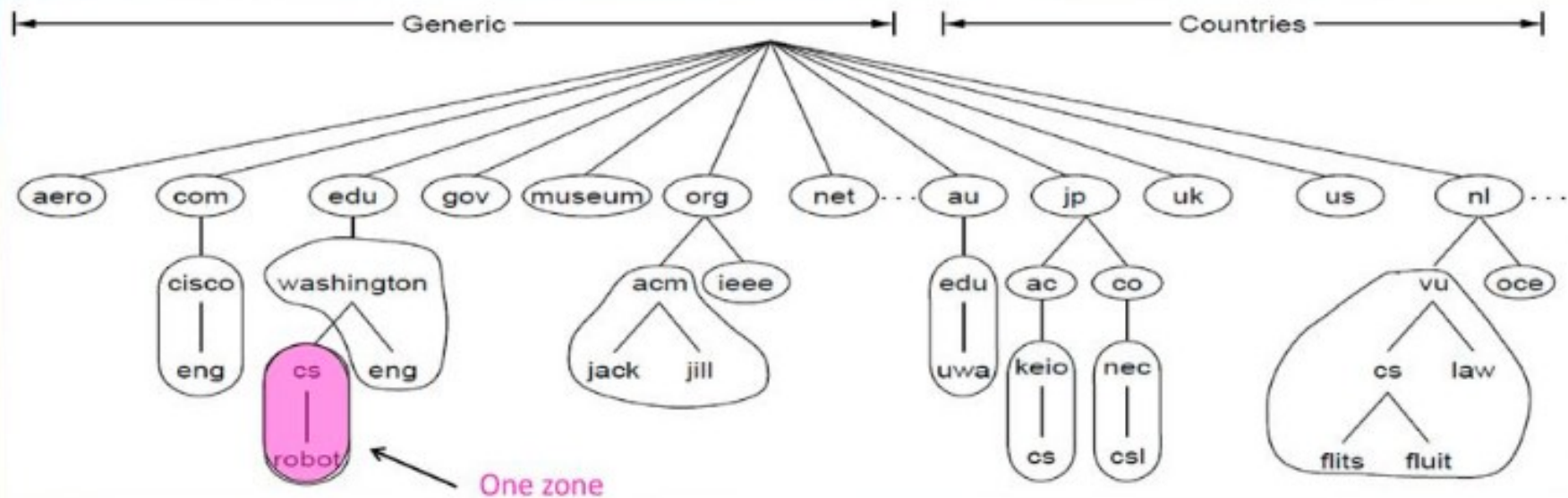
Domain Resource Records

The key resource records in the namespace are IP addresses (A/AAAA) and name servers (NS), but there are others too (e.g., MX)

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

Name Servers

Name servers contain data for portions of the name space called zones (circled).



Name Servers

Finding the IP address for a given hostname is called resolution and is done with the DNS protocol. (Remember ARP?)

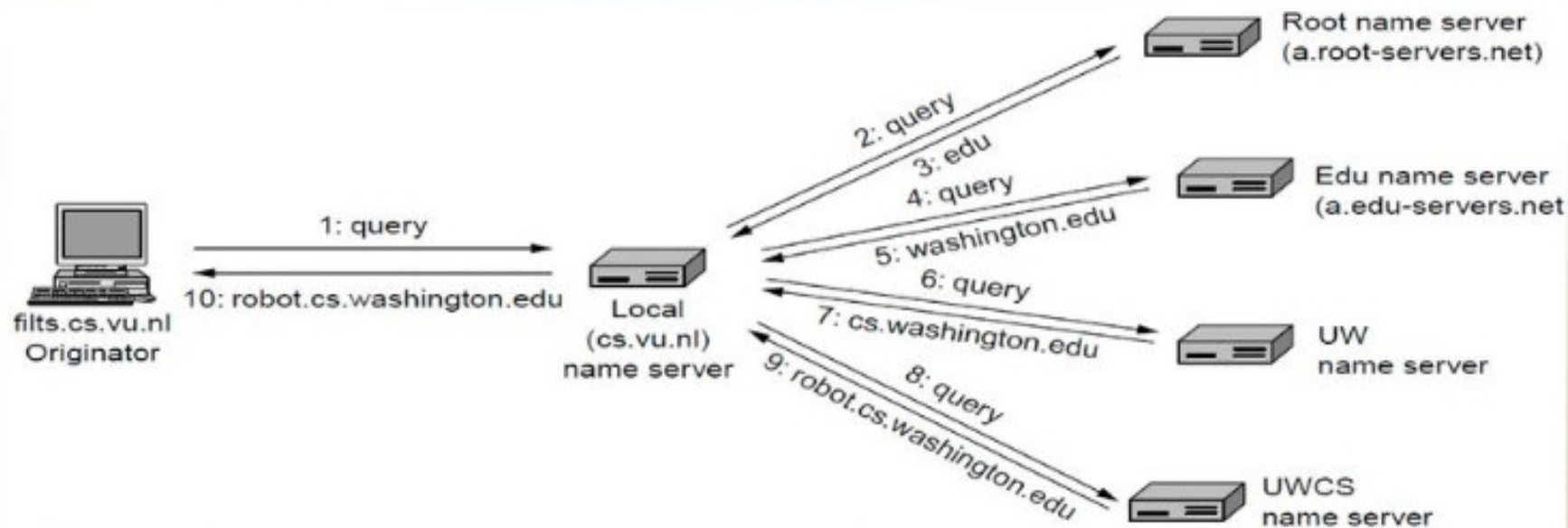
Resolution:

- Computer requests local name server to resolve
- Local name server asks the root name server
- Root returns the name server for a lower zone
- Continue down zones until name server can answer

DNS protocol:

- Runs on UDP port 53, retransmits lost messages
- Caches name server answers for better performance

Name Servers



Example of a computer looking up the IP for a name

Electronic Mail

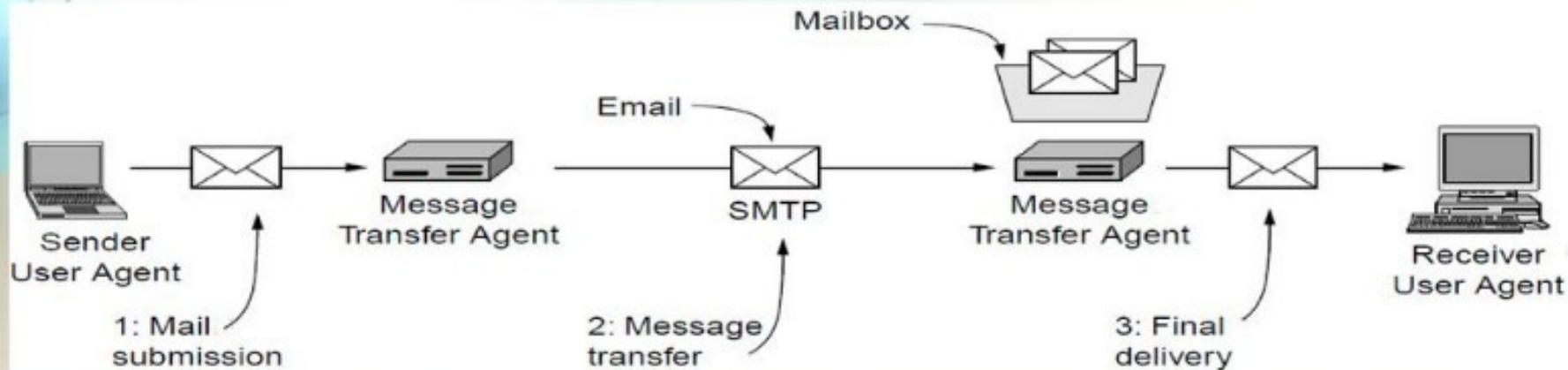
- Architecture and services »
- The user agent »
- Message formats »
- Message transfer »
- Final delivery »

Electronic Mail

- The architecture of the email system consists of two kinds of subsystems:
 - The **user agents**, which allow people to read and send email and
 - The **message transfer** agents, which move messages from the source to the destination.
- **User agents** or email readers are programs such as Google gmail. Microsoft Outlook, are Apple Mail that accepts commands for composing receiving, editing and replying to messages.
- **Message transfer agents** are usually system processes, which run in the background on mail servers. They use SMTP (Simple Mail Transfer Protocol) to transfer messages.

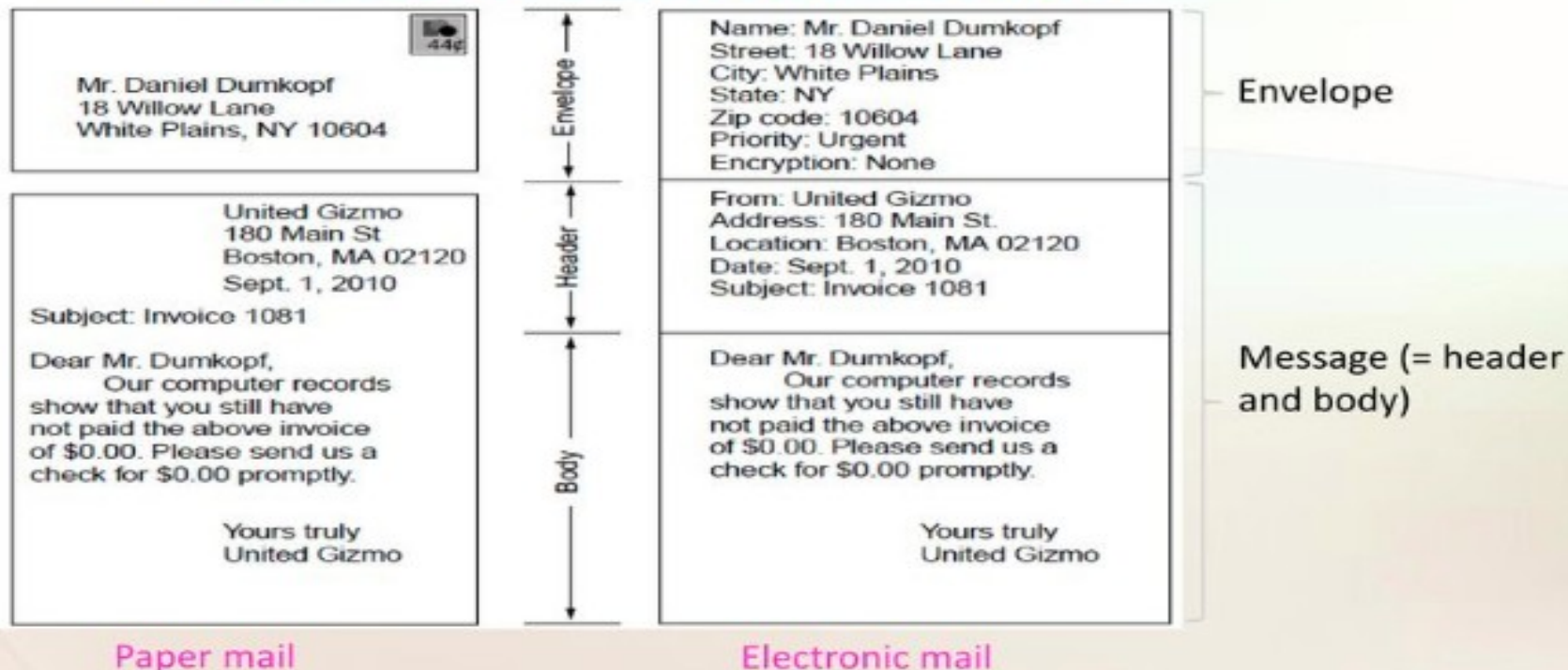
Architecture and Services

- The key components and steps (numbered) to send email



Architecture of the email system

Architecture and Services



The User Agent

Message folders

Mail Folders
All items
Inbox
Networks
Travel
Junk Mail

Search 

Mailbox search

Message summary

From		Subject	Received
trudy		Not all Trudys are nasty	Today
Andy		Material on RFID privacy	Today
djw	!	Have you seen this?	Mar 4
Amy N. Wong		Request for information	Mar 3
guido		Re: Paper acceptance	Mar 3
lazowska		More on that	Mar 2
lazowska		New report out	Mar 2
...		...	...

A. Student	Graduate studies?	Mar 1
Dear Professor, I recently completed my undergraduate studies with distinction at an excellent university. I will be visiting your ...		

Message

Message Formats

Header fields related to message transport; headers are readable ASCII text

Header	Meaning
To:	Email address(es) of primary recipient(s)
Cc:	Email address(es) of secondary recipient(s)
Bcc:	Email address(es) for blind carbon copies
From:	Person or people who created the message
Sender:	Email address of the actual sender
Received:	Line added by each transfer agent along the route
Return-Path:	Can be used to identify a path back to the sender

Message Formats

- Basic ASCII Formats
- RFC 822 – original format – did not distinguish envelope fields from header fields
- RFC 5233- Internet Message Format used in SMTP
- MIME (Multipurpose Internet Mail Extension) – 1990's – added structure and defined encoding rules for non-ASCII messages to allow for audio, images and binary files.

Message Formats

Other header fields useful for user agents

Header	Meaning
Date:	The date and time the message was sent
Reply-To:	Email address to which replies should be sent
Message-Id:	Unique number for referencing this message later
In-Reply-To:	Message-Id of the message to which this is a reply
References:	Other relevant Message-Ids
Keywords:	User-chosen keywords
Subject:	Short summary of the message for the one-line display

Message Formats

MIME header fields used to describe what content is in the body of the message

Header	Meaning
MIME-Version:	Identifies the MIME version
Content-Description:	Human-readable string telling what is in the message
Content-Id:	Unique identifier
Content-Transfer-Encoding:	How the body is wrapped for transmission
Content-Type:	Type and format of the content

Message Formats

Common MIME content types and subtypes

Type	Example subtypes	Description
text	plain, html, xml, css	Text in various formats
image	gif, jpeg, tiff	Pictures
audio	basic, mpeg, mp4	Sounds
video	mpeg, mp4, quicktime	Movies
model	vrml	3D model
application	octet-stream, pdf, javascript, zip	Data produced by applications
message	http, rfc822	Encapsulated message
multipart	mixed, alternative, parallel, digest	Combination of multiple types

Message Transfer

Messages are transferred with SMTP (Simple Mail Transfer Protocol)

- Readable text commands
- Submission from user agent to MTA on port 587
- One MTA to the next MTA on port 25
- Other protocols for final delivery (IMAP, POP3)

Message Transfer

Sending a message:

- From Alice to Bob
- SMTP commands are marked [pink]

```
[ ] S: 220 ee.uwa.edu.au SMTP service ready
C: HELO abcd.com S: 250 cs.washington.edu says hello to ee.uwa.edu.au
[ ] C: MAIL FROM: <alice@cs.washington.edu> S: 250 sender ok
[ ] C: RCPT TO: <bob@ee.uwa.edu.au> S: 250 recipient ok
C: DATA S: 354 Send mail; end with "." on a line by itself
C: From: alice@cs.washington.edu
C: To: bob@ee.uwa.edu.au
C: MIME-Version: 1.0
C: Message-Id: <0704760941.AA00747@ee.uwa.edu.au>
C: Content-Type: multipart/alternative; boundary=qwertyuiopasdfghjklzxcvbnm
C: Subject: Earth orbits sun integral number of times
C:
C: This is the preamble. The user agent ignores it. Have a nice day.
C:
C: --qwertyuiopasdfghjklzxcvbnm
C: Content-Type: text/html
C:
C: <p>Happy birthday to you
C: Happy birthday to you
... (rest of message) ...
C: --qwertyuiopasdfghjklzxcvbnm
C: .
[ ] C: QUIT S: 250 message accepted
S: 221 ee.uwa.edu.au closing connection
```

Final Delivery Protocols

- SMTP is not used for final delivery- it takes a message and connects to a remote server to transfer the message. It is a “push-based” protocol. (The user agent may not be connected to the Internet when SMTP attempts to deliver the message.)
- IMAP- Internet Message Access Protocol is used for final delivery. The IMAP server listens to port 143 and is an improvement over older protocols
- POP3 – Post Office Protocol – version 3 - fewer features- less secure
- Webmail – interface for sending and receiving email (gmail, hotmail, yahoo) – an example of software provided as a service (SaaS) using the web.

POP3

It allows user transfer agents (on client PCs) to contact the message transfer agent (on the ISP's machine) and allow e-mail to be copied from the ISP to the user.

POP3 is an abbreviation for Post Office Protocol, whereas '3' refers to the version. Version 3 is the latest and most widely used one.

It is a simple protocol not having much to offer apart from the download.

POP3 downloads an email from the server to one computer (or device) and ends up removing it from the server once it gets downloaded on your computer. That means it won't be possible to read your email messages from multiple locations, which is not ideal.

The best part of this protocol are that you can read your emails when you are offline. And the use of the email server's storage is less.

However, as it works with one device and doesn't store the messages on the server, there is a need to backup your computer. Failing to do so will result in you losing all your emails when your computer dies.

Working of PoP3

There are four stages in a POP3 connection,

1. Authorization stage – The client connects with the server.
2. Transaction stage – The client retrieves the email
3. Update Stage – The server deletes the message stored
4. The client and server disconnect from each other

Advantages of POP3

- Offline access: As you just read, POP3 allows users to access their email messages even when they are offline. This is because the email messages are downloaded to the user's device and can be read and replied to at any time.
- Simple to set up: POP3 is easy to set up and use, making it a popular choice for many email users. It only requires a few settings, such as the user's email address and password, to be entered into the email client.
- Efficient use of bandwidth: POP3 is a bandwidth-friendly protocol, as it only downloads email messages when the user requests them. This means that it does not use unnecessary bandwidth, which is particularly useful for users with limited internet connectivity.
- Lower resource requirements: POP3 has lower resource requirements than other email protocols, such as IMAP. This means that it works on less powerful hardware or in environments where resources are limited.

IMAP

- IMAP is an abbreviation for Internet Message Access Protocol. It is a popular protocol for receiving email messages. The most significant advantage of the IMAP protocol is that it lets you receive email on more than one computer (or device). That is because after getting delivered, the email stays on the mail server. In short, you can read your email on your office computer, the desktop at home, your tablet and your smartphone.
- Also, it doesn't download the entire message until you open it. That helps in a faster initial connection and startup.
- However, it won't perform well when you have a slow internet connection. And you won't be able to read your emails.

- The basic IMAP client and server interaction is as mentioned below,
 1. The email client of the recipient connects with the server where the message is stored.
 2. The recipient can view the message headers on the server.
 3. If the recipient selects a message to read it, IMAP downloads that one for the recipient.

Advantages of IMAP

- Access to emails from multiple devices: As you just read above, IMAP allows users to access their email messages from multiple devices. The email messages are stored on a server, not on the user's device.
- Synchronization: IMAP synchronizes the user's email messages across all devices. This means that any changes made to an email message on one device automatically reflect on all other devices.
- Offline access: IMAP allows users to access their email messages even when they are offline. This is because a copy of the email messages is stored on the user's device, and changes are synchronized with the server when the user goes online.
- Efficient use of storage space: IMAP allows users to manage their email messages more efficiently. This is because only the header of an email message is downloaded to the user's device, and the full message gets downloaded when the user opens it.

Key	POP3	IMAP
Full Form	POP3 stands for Post Office Protocol 3	IMAP stands for Internet Message Access Protocol
Complexity	POP3 is simple and only mails can be downloaded from your inbox to local computer.	IMAP is complex and allows to see all the folders on the mail server.
Ports	POP3 listens on 110 and POP with SSL, POP3DS listens on 995 port.	IMAP listens on 143 and IMAP with SSL, IMAPDS listens on 993 port
Multiaccess	POP3 supports single device to access the mail at a time.	IMAP supports multiple devices which can access the mail at a time.
Download	In POP3, mail to be downloaded first then can be read.	In IMAP, mail can be partially read before complete download.
Mail Organize	Mails cannot be organized on mail server using POP3.	IMAP allows to organize mails on mail server.
Update Email	Mails cannot be created/updated/deleted on mail server using POP3.	IMAP allows to create/update/delete mails on mail server.
Search Content	Mail content cannot be searched on mail server using POP3. To search, mail to be downloaded first.	Mail content can be searched on mail server using IMAP.
Download	All message are downloaded at once.	Mail message header can be previewed before a message is to be downloaded.

HTTP—The HyperText Transfer Protocol

The transfer protocol used throughout the World Wide Web is HTTP (HyperText Transfer

Protocol). It specifies what messages clients may send to servers and what responses they

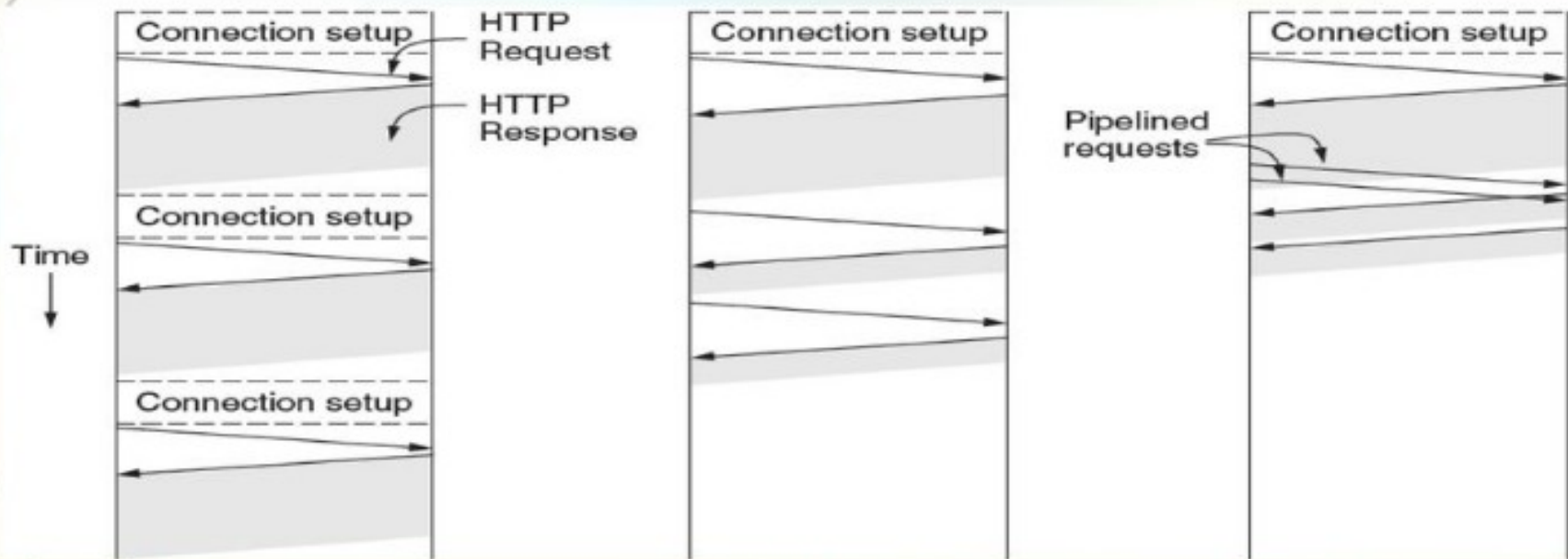
get back in return. Each interaction consists of one ASCII request, followed by one RFC 822

MIME-like response. All clients and all servers must obey this protocol. It is defined in RFC

2616.

HTTP (HyperText Transfer Protocol) is a request-response protocol that runs on top of TCP

- Fetches pages from server to client
- Server usually runs on port 80
- Headers are given in readable ASCII
- Content is described with MIME types
- Protocol has support for pipelining requests
- Protocol has support for caching



One connection for each request

Sequential requests on one connection

Pipelined requests on one connection

Connections

- The usual way for a browser to contact a server is to establish a TCP connection to port 80 on the server's machine,
- In HTTP 1.0, after the connection was established, a single request was sent over and a single response was sent back. Then the TCP connection was released. Within a few years, the average Web page contained large numbers of icons, images, and other eye candy, so establishing a TCP connection to transport a single icon became a very expensive way to operate.
- This observation led to HTTP 1.1, which supports persistent connections. With them, it is possible to establish a TCP connection, send a request and get a response, and then send additional requests and get additional responses.

Methods

HTTP has several request methods.

Fetch a page



Used to send input data
to a server program



Method	Description
GET	Read a Web page
HEAD	Read a Web page's header
POST	Append to a Web page
PUT	Store a Web page
DELETE	Remove the Web page
TRACE	Echo the incoming request
CONNECT	Connect through a proxy
OPTIONS	Query options for a page